

Mounting Solutions for Home Network Equipment

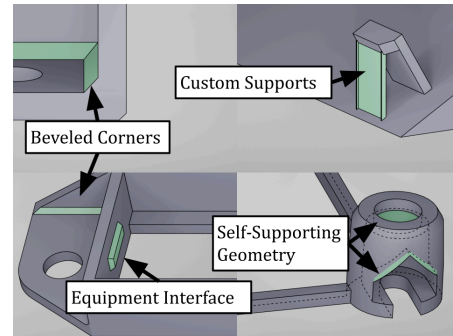
Hardware Prototyper / Apr 2025

Overview:

Designed and 3D printed custom mounting brackets to efficiently reorganize home network equipment. The project emphasized compact, strong, and easily installable parts, with careful attention to printability through optimized hole geometries and part orientation. Rapid prototyping enabled quick fit testing and iterative improvements.

Tools & Skills:

Blender, 3D Printing, Optimization for 3D Printing, Mechanical Design, Rapid Prototyping



Key structural and print-optimized features across mount designs.

Objective:

Designed and 3D printed a series of custom mounts and brackets to reorganize home network equipment. Focused on space efficiency and ease of installation.

Design Highlights:

- Developed multiple custom-fit mounting brackets tailored to various devices.
- Designed compact, functional parts optimized for strength and print efficiency.
- Applied printability optimizations to hole geometries, including teardrop shapes and thin membranes on overhangs, to ensure reliable printing.

Reflection:

Rapid prototyping by 3D printing small sections for fit testing and interfacing with other components enabled efficient design iteration and verification. Designing for 3D fabrication required careful planning of part orientation and intentional hole geometry to ensure successful printing.